

Cash Flow Minimization Report

1. Group Members

Name	Email ID
lokesh	lokeshwarvarma09@gmail.com
gowtham	moggowtham@gmail.com
monish	valivetimonish@gmail.com

2. Registered Expenses

cafe (Total: Rs. 530.65) - Paid by lokesh

-> gowtham owes lokesh Rs. 176.88

-> monish owes lokesh Rs. 176.88

dineer (Total: Rs. 853.23) - Paid by gowtham

-> lokesh owes gowtham Rs. 230.00

3. Initial Debt Network (Before Minimization)

Payer (Owes)	Payee (Receives)	Amount
gowtham	lokesh	Rs. 176.88
monish	lokesh	Rs. 176.88
lokesh	gowtham	Rs. 230.00

4. Minimized Debts (Greedy Algorithm)

-> monish must pay lokesh : Rs. 123.77

-> monish must pay gowtham : Rs. 53.12

5. Minimized Debts (Graph Flow Algorithm)

Accounts are fully balanced. No transactions needed.

Optimization Summary: Greedy reduced 1 edges. Flow reduced 3 edges.

Analytics & Visualizations

Greedy Algorithm Network:



